

StainlessCarbon

Research Appendix

The evidence base behind the carbon and energy calculator, the Round 1 summary and the Round 2 deck: emission factors, methodology, regulation, company disclosures, competitor analysis, and the locked canonical model (v4) that every submitted figure traces to.

Live calculator: grv-io.github.io/sc-346a9f98/website
Date: 11 September 2026, updated 26 September 2026 for Round 2 (canonical model v4)

Independent case-competition project built on public disclosures, not an official Jindal Stainless product. Figures carry the uncertainties noted per section.

Executive summary

The model in three lines. Scope 1 = costed captive coal power + own-furnace ferrochrome reductant + fired fuel + AOD. Scope 2 = purchased power only, after captive, times the renewable share. Scope 3 = merchant ferrochrome plus nickel, FeMo, FeMn, copper and consumables, capped by grade-specific scrap limits. The CBAM figure strips captive-power carbon back out and adds only purchased precursors' direct emissions, per installation, per tonne of coil.

Check	Target	Model
FY26 company Scope 1+2 (fit)	1.76	1.760
FY26 company total, as disclosed (fit)	3.03	3.031
FY24 Scope 1+2, out-of-sample	2.15 ±2%	2.164 (+0.65%)
FY24 Scope 1, out-of-sample	1.702 ±3%	1.716 (+0.8%)

The five things the problem statement asked us to show, in one line each.

- Reasonableness of the model:* every constant is tagged JSL-assured, regulation, literature, derived, assumption or calibration-fit; FY24 lands within 0.8% out-of-sample and 415 automated checks pass.
- Clarity of how inputs affect the result:* five levers plus grade drive a Shapley-attributed breakdown of each input's marginal effect on Scope 1, 2, 3 and the CBAM figure separately, not one total.
- An interactive, easy-to-understand interface:* a free, phone-usable calculator with five sliders and a grade choice, live at the link above.
- Usefulness of the optimisation/comparison feature:* three cost-tiered optimiser paths (today, 2030, theoretical), each with disclosed \$/tCO2 costs, plus a route comparison across Jajpur, Hisar and an Indonesian slab route.
- A measurable reduction in estimated emissions:* FY26's 3.03 t/t falls to 2.30 t/t on levers buildable today, and to 0.73 t/t at the theoretical floor.

3.03 → 2.30
t CO2e/t, FY26 disclosed total vs the practical-tier optimiser (buildable today)

1.17/t coil
Jajpur's verified CBAM figure, 2026, per tonne of cold-rolled coil

≈₹139 Cr/yr
net to JSL once Iberjindal volume, pricing capture and verification cost are worked

1. The canonical model: locked figures and rulings

Full text: [research/22-canonical-model-v1.md](#) (v4; filename kept for link stability). Executable form: [website/model.js](#) (415/415 tests) and [website/business.js](#) (110/110 tests).

Pre-submission audits found conflicts across the research base, resolved by written ruling; the rulings, not the underlying files, govern all submitted numbers. v4 splits captive/merchant ferrochrome sourcing, prices the captive load factor as a real lever, stops crediting sourcing changes that relocate emissions as cuts, adds three optimiser tiers, separates the disclosed total from a true cradle-to-gate estimate, and puts CBAM's own quota into the money the tool reports.

- **CBAM dataset (Ruling 1).** India default **6.49 tCO₂e/t** base, **7.14 / 7.79 / 8.44** with mark-up; price **€75** in 2026, then €89/95/100/109 (analyst path); phase-in 2.5% to 48.5%. EU stainless-flat quota **64,073 t/yr for all Indian mills**; out-of-quota duty **50%**.
- **CBAM figure (Ruling 1b).** Direct emissions only: captive-power carbon stripped out, purchased precursors' direct emissions added in. Renewables and the captive load factor move it by zero; grade does. v4 adds verified vs EU-default precursor factors (both flagged against the regulation's Annex) and a per-coil basis (now the default).
- **The money that reaches JSL (Ruling 1d, new).** The buyer-side saving is not automatically JSL's; any verifier closes that gap. On quota-feasible tonnes, JSL's own value is pricing capture on third-party volume plus Iberjindal's own volume as importer of record, minus verification cost and out-of-quota duty (section 6).
- **Per installation (Ruling 1c).** Jajpur verified **1.171/t coil** (conservative 2.418); Hisar **0.887** (conservative 0.798); Indonesia-slab route **9.429**. The company average (1.76) is filed by nobody.
- **Captive power, a costed lever (Ruling 11).** 264 MW Jajpur × load factor **0.70** (range 0.55 to 0.80), baseload; renewables apply to purchased power only; priced (₹2 to 3/kWh) and in every ranking, not a what-if.
- **Own-furnace ferrochrome: site attribution, partial split (Ruling 12).** Own-SAF FeCr under the site rule is **4.92 t/t**, close to merchant Odisha's 5.65, since it now carries the plant's real electricity mix; the captive share of chromium units is partial and capacity-limited, not all-or-nothing.
- **Relocation excluded from progress (Ruling 12b, new).** Sourcing switches into Scope 3 are excluded from the FY35 progress figure and the optimiser's targets, not just noted.
- **Footprint labelling and the customer figure (Ruling 14, new).** The 3.031 total is "as disclosed" (includes downstream categories 10, 12), never "cradle-to-gate"; the true estimate is 2.911. The customer figure follows the installation, per t coil (Jajpur 3.58, Hisar 2.59), never the company blend.
- **An Indonesia melt-shop route (Ruling 15, new).** Indonesian slab, rolled in India, prices worse on the EU default than India's own default (9.43 vs 7.14/t coil); a declarant would currently file the Indian default.
- **Optimiser tiers and real costs (Ruling 16, new).** Three tiers (today, stretch 2030, theoretical), rising disclosed \$/tCO₂ steps; hydrogen priced separately from biochar (\$250 vs \$29 to 45/tCO₂).
- **Benchmarks.** Never compare S1+2 (1.76) with Outokumpu's 1.8 (cradle-to-gate incl. Scope 3). Use: total as disclosed 3.03 vs worldstainless approximately 3.70; S1+2 1.76 vs Indian steel-sector approximately 2.5.

1a. What changed since Round 1

Round 1's summary used a gross pro-rata CBAM shortcut and an early, less disaggregated model. This table reconciles the two rather than letting the old figures stand.

Round 1 said	Round 2 says	Why
₹200–290 Cr/yr by 2030	Buyer-side ₹135–161 Cr in 2026 at 40 kt; ₹265–285 Cr by 2030; ≈₹139 Cr/yr net to JSL	The net formula replaces the pro-rata shortcut, so the saving exists from 2026; the headline now separates what any verifier gets from what lands with JSL
€372–398/t (2030)	€749–805/t crude in 2030 (€736–792/t coil); €374–448/t coil in 2026	EC guidance: Liability = (embedded minus benchmark × free-allocation) × price, not embedded × phase-in × price
Calibration: S1+2 1.73, total 2.99	S1+2 1.760, total 3.031	Re-fitted to the FY26 Integrated Report per installation
FY28–30 path: S1+2 1.35–1.45	S1+2 1.52 on the FY30 roadmap preset	Captive coal is now baseload; renewables only replace purchased power, so levers cut less
"430 is about half of 304"	430 / 304 = 0.62	430 still carries the same ferrochrome and melt energy; ratio is JSL-specific, not the global average
"60–80% of the footprint is alloys"	36% bought-in nickel and ferro-chrome; 51% including own-furnace ferro-chrome	60–80% holds on low-scrap routes; JSL's 70% scrap makes the virgin charge smaller
"No existing calculator does this"	Aperam and Outokumpu publish producer tools; neither models an Indian mill's alloy chain per installation on the CBAM boundary	the differentiator is the alloy chain plus the per-installation figure, not being first
An earlier, unofficial product tagline	StainlessCarbon	One name across calculator, appendix and deck

2. Emissions model specification and calibration

S1 = captive power [min(Cmax, E_total)·1.045, Cmax = 264MW·8760h·LF / crude] + SAF reductant [safOut·1.5·(1-biochar)] + fired fuel [F0·(1-cleanfuel), F0 = 0.531 Jajpur / 0.270 Hisar] + AOD [0.0897 + 44/12·C] S2 = purchased·(1-RE)·0.710, purchased = E_total minus captive (RE applies to purchased power only) S3 = need·(1-X)·FeCr(merchant, X = captive share of Cr units) + Ni units + FeMo + FeMn + Cu + Fe-units + consumables(0.18) CBAM E = S1 minus captive-power CO2 minus SAF-for-precursor CO2 + purchased-precursor direct emissions (FeCr/NPI/FeMn/pig iron), ÷ yield 0.90 for the per-coil figure subject to grade-specific scrap caps: 304 80% / 316 75% / 430 60% / 2205 60% / J4 70%
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Calibration constants: fired fuel Jajpur **0.531**, Hisar **0.270** t/tcs; AOD **0.0897** t/t plus a carbon mass balance; electricity **1.428 MWh/tcs** (captive 0.803, purchased 0.626, SAF 0.356); thermal Jajpur **5.958** / Hisar **5.440** GJ/tcs; NPI share **58%** (a plug); clean fuel **2%** (derived, not invented). Captive power is baseload at load factor 0.70 (costed, range 0.55 to 0.80); own-furnace ferrochrome takes the plant's real electricity mix by default (a marginal-supply alternative exists for comparison), giving own-SAF FeCr of 4.92 t/t against merchant Odisha's 5.65, with a partial captive/merchant split, not all-or-nothing.

Energy output: GJ per tonne crude, anchored to JSL's disclosed **16.43 GJ/tcs** (FY26; FY24 19.97).

Grade-wise virgin-charge chemistry (kg per tonne crude steel)

Grade	Cr %	Ni %	Mo %	HC FeCr	Contained Ni	FeMo	HC-FeMn	Cu	Scrap cap
304 austenitic	18	8	–	300	80	–	–	–	80%
316 Mo-grade	17	10	2	283	100	29	–	–	75%
430 ferritic	17	–	–	283	–	–	–	–	60%
2205 duplex	22	5	3	367	50	43	–	–	60%
J4 (200-series)	15	1	–	250	10	–	120	16	70%

Validation, stated honestly. (1) <i>FY26 anchor, identity not proof:</i> S1+2 1.760, total 3.031 vs audited 1.76/3.03, within 0.1%; seven fitted constants reproduce seven targets exactly. (2) <i>FY24 out-of-sample, the real test:</i> S1+2 2.164 vs 2.15, within 0.65%; Scope 1 within 0.8%. (3) <i>Industry curve:</i> within ±15% of worldstainless at 30% scrap (6.913 vs 6.80) and 50% scrap (3.714 vs
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3.70). (4) *Grade physics*: 430 is 0.62 of 304's total, not the ISSF ratio of approximately 0.95, disclosed as a difference, not a match.

3. Emission factors: routes, alloys, India fuels and power

Files 01, 04, 05. Sources: worldsteel, Columbia CGEP, ICDA, Metso Outotec, ISSF, CEA v21.0, BEE PAT, IPCC 2006, MoEFCC BUR-3

Steelmaking routes (cradle-to-gate unless noted)

Route	tCO ₂ /t crude steel
BF-BOF (virgin)	~2.2 (1.8–3.0)
Scrap-EAF	~0.67
DRI-EAF (natural gas)	0.7–1.2
DRI-EAF (coal)	2–3
EU CBAM plant-gate benchmarks (reference)	BF-BOF 1.370 · DRI-EAF 0.481 · scrap-EAF 0.072

Alloy chain: the stainless differentiator

Input	Factor	Status	Note
HC ferrochrome, global average (imported option)	2.3 tCO ₂ /t	literature	closed SAF; ICDA range 2.3–2.8
HC ferrochrome, Odisha captive-coal (merchant share)	5.4–5.9 (tool 5.65)	literature	bought-in share only; JSL data ask
Own-furnace ferrochrome, site electricity mix	4.92 t/t (1.5 direct + 3.95·0.865)	literature/calibration	inside Scope 1+2; marginal-supply alt. 2.99 t/t
NPI, coal power / RE-hydro (New Yaking JV)	69 / 30 t/t Ni	literature / assumption	69 central (60–85); 30 = JV on RE/hydro
Class-1 nickel / ferromolybdenum	~13 / 6.5 ±30%	literature	not CBAM goods; FeMo least-certain
HC-FeMn / copper / pig iron (J4)	3.5 (1.3 direct) / 4.0 / 2.1 (1.9 direct)	literature/assumption	FeMn, pig iron are CBAM precursors; copper is not
Consumables	0.18 tCO ₂ /tcs	assumption	range 0.12–0.27
CBAM precursor defaults (EU regulation)	FeCr 2.35 · NPI 6.15/t NPI (50.4/t Ni) · FeMn 1.69 · pig iron 2.53	regulation	third-party compilation; verify against the Annex

India power and fuels (regulator-grade)

Source	Factor	Status	Basis
Grid electricity (FY24-25)	0.710 tCO ₂ /MWh	literature	CEA CO ₂ Baseline Database v21.0
Captive subcritical coal power	1.045 tCO ₂ /MWh	literature	CEA default heat rate 2,500 kcal/kWh; range 1.04–1.15
Captive plant capacity and load factor	264 MW × LF 0.70	literature (capacity); LF assumption	264 MW per company statements; LF now a costed lever, range 0.55–0.80
Domestic coal (G12 blend)	1.46 tCO ₂ /t	literature	BEE PAT / CEA App. B
LSHS / furnace oil	3.16 / 3.04 tCO ₂ /t	literature	BEE / CEA
Natural gas	1.82 kg/SCM (–31% vs FO in reheating)	literature	CEA App. B
Bio-LDO / green hydrogen	≈0 fossil Scope 1	literature	IPCC biogenic accounting

4. Methodology alignment: worldstainless / ISSF

File 06. ISO 14040/14044, ISO 14404, GHG Protocol, worldsteel LCI

- Standard boundary: cradle-to-gate (upstream alloys = Scope 3; melting/refining and downstream = Scope 1+2). Upstream Scope 3 is 60–75% of the footprint on virgin/NPI routes; at JSL's own 70% scrap it is smaller (36 to 51% of the FY26 total, see section 1a).
- Recycled-content method (cut-off): scrap enters burden-free; virgin alloys carry full upstream burden, used for factory-gate reporting and CBAM. End-of-life method (IISI Appendix 5) is implemented as a second output.
- Benchmark curve (worldstainless): 85% scrap → 1.95 · 75% → 2.45 · 50% → 3.70 · 30% → 6.80 tCO₂/t. Rule of thumb: +10pp scrap is approximately –0.8 to –0.9 tCO₂/t on a virgin-heavy route; JSL's own slope is shallower (–0.54) because 42% of its nickel is already Class-1 and own-SAF ferrochrome sits at 4.92 t/t, not 5.65.

5. Jindal Stainless: disclosed emissions FY22–FY26

Files 03, 16. JSL ESG Factsheets, BRSR, CDP, Integrated Annual Reports incl. FY26 Integrated Report (Aug 2026)

Metric (tCO ₂ e/tcs)	FY22	FY23	FY24	FY25	FY26
Scope 1+2 intensity	1.98	2.08	2.15	1.85	1.76
Scope 3 intensity	n/r	1.76	1.90	1.64	1.27*
Total, as disclosed	–	–	4.05	3.49	3.03*
Energy intensity (GJ/tcs)	–	–	19.97	–	16.43

* FY26 Scope 3 = 2,561,950 t; the IR footnote attributes part of the drop to a method change for categories 10 and 12. This total includes those categories, so it is JSL's disclosed boundary, not cradle-to-gate (section 1).

- FY26 Scope 1 = 3,074,950 t; Scope 2 = 474,893 t market-based / 651,984 t location-based; S1+2 (market) 3,549,843 t ÷ 1.76 = 2.017 Mt crude.
- Per installation (FY26 S1+2): Jajpur **2.31** (Scope 1 2,803,152 t, 91% of company), Hisar **0.74**, company 1.76.
- Levers on record: 70.12% scrap; renewables 46.8% of imported power (315.6 MW hybrid); green hydrogen 90/600 Nm³/h Hisar/Jajpur; Bio-LDO 30% of Hisar hot-rolling fuel; FY35 target 0.99; Net Zero 2050.

6. CBAM: the statutory formula, the quota and the money

Canonical model v4 · EC CBAM Q&A (27 May 2026) §3.8 · Regulation (EU) 2023/956; Implementing Regulations (EU) 2025/2620, 2025/2621, 2026/1740; EU steel safeguard 2026 revision · EC CBAM certificate price page

Section 1 (rulings) and section 1a (reconciliation) set out the definition and the per-installation figures; this section carries the worked arithmetic those bullets summarise.

Liability €/t = max(0, E minus BM × f) × P f = free-allocation share 97.5 / 95 / 90 / 77.5 / 51.5% (2026–30)
 BM = CBAM benchmark: Column A if verified emissions are declared, Column B if defaults are used · base case A = B = 0.268

Year	P €	f	Default E	Default €/t	Jajpur verified 1.171 €/t coil	Saving, verified	Saving, conservative
2026	75	0.975	7.14	515.9	68.2	447.7	375.1
2027	89	0.95	7.79	670.7	81.6	589.1	503.0
2028	95	0.90	8.44	778.9	88.3	690.6	598.5
2029	100	0.775	8.44	823.2	96.3	726.9	629.9
2030	109	0.515	8.44	904.9	112.4	792.3	687.4

2026 row: default (7.14 minus 0.2613) × 75 = 515.9; verified (1.171 minus 0.2613) × 75 = 68.2; saving = 447.7. Conservative row uses Jajpur's Scope 1 per precursors (2.418). Per t crude (not per coil), the 2030 saving range across the unverified default-side benchmark is €749–805/t.

Who actually captures it. Third-party buyers pass through roughly 40% of their avoided cost (≈₹14 Cr/yr); Iberjindal S.L., assumed importer of record for its approximately 30 kt/yr, captures its saving whole (≈₹127–128 Cr/yr if confirmed); less verification cost (≈₹0.8 Cr/yr) and out-of-quota duty (₹0, base case: those tonnes are diverted rather than shipped at 50% duty). **Net to JSL ≈ ₹139 crore a year**, computed live, contingent on confirming Iberjindal's importer-of-record status.

Quota, routing and sensitivities

Item	Value
India stainless-flat EU quota (all mills)	64,073 t/yr (CR 38,054 + HR 26,019); 50% out-of-quota duty
JSL share / in-quota at 40 kt exports	60% / 38,444 t in-quota, 1,556 t out-of-quota
Landed cost, verified vs default	€3,127 vs €3,574 (CBAM 2.2% vs 14.4% of landed)
Hisar vs Jajpur routing	Hisar €21/t coil cheaper on CBAM
Indonesian-slab coil vs Jajpur-melted	+€619/t coil worse, filed on the EU default
Column B 0.268 / 0.6 / 0.9 / 1.27	2026 saving €447.7 / 423 / 401 / 374.4 per t coil
Captive load factor 0.55 / 0.70 / 0.80	No effect on the CBAM figure (electricity is excluded)

Uncertainties. Column B is unverified; India's default may be 5.59, not 6.49; the quota's regulation number is unresolved (2026/1457 vs 2026/1384). Floor case (both together): 2026 saving falls to approximately €309/t crude. Dec-26 EUA (≈€86) is a 2026 upside only.

Iberjindal and timing. Iberjindal S.L. (Lucena, Spain), JSL's wholly owned subsidiary, is assumed sourced from Hisar; if it is importer of record its saving lands in JSL's consolidated results. Melt-and-pour origin (from 1 Oct 2026) fixes CBAM origin at the melting furnace. The first CBAM declaration for 2026 imports is due 30 September 2027, so 2026 tonnes can still switch default to verified.

7. EU market access beyond CBAM: CPR, ESPR, Digital Product Passport

Files 12, 20. Regulation (EU) 2024/3110 (CPR), 2024/1781 (ESPR), CEN/CENELEC EN 18216–18223

- CPR: from 8 Jan 2026, EU construction-market steel must declare facility-specific GWP (EN 15804+A2) for CE marking.
- ESPR/DPP: iron and steel are first-wave priority products, mandatory compliance approximately 2028; the Digital Product Passport is a decentralised, machine-readable record.
- Green procurement: the proposed Industrial Accelerator Act would require at least 25% low-carbon steel in EU public procurement from 2029.
- The calculator's per-tonne record feeds EPD/DoPC and DPP data fields directly: compliance infrastructure, not just an estimator.

8. Jindal's EU exposure: exports, quotas, competitive position

File 19. CARE Ratings, JSL results, Iberjindal disclosures, UN Comtrade, Goldman Sachs / ICRA / CRISIL

- Sales volume 2.17 to 2.56 Mt (FY24 to FY26); export mix moderated 17% to 8%; estimated direct EU shipments 60–80 kt/yr against India's 64,073 t quota, approximately 30 kt through Iberjindal S.L.
- Mid-2026 safeguard revision cut the tariff-free quota 47% and doubled the out-of-quota duty to 50%; CRISIL sees a 25–30% decline in Indian steel exports to the EU. Goldman Sachs: Indian BF-BOF steel faces \$102–190/t CBAM cost; JSL's scrap-EAF route with verified data is the structural exception.

Six verified business facts

- **The EU quota is binding, not abundant:** India's tariff-free quota is CR 38,054 + HR 26,019 = **64,073 t/yr for all Indian mills**; out-of-quota tonnes carry a 50% duty. JSL's own 60–80 kt estimate alone would fill it, so the case is value per quota tonne, not volume. *Regulation number to be confirmed: 2026/1457 or 2026/1384.*
- **Melt-and-pour origin from 1 Oct 2026** fixes CBAM origin at the melting furnace, so Jajpur- or Hisar-melted product carries JSL's own figure into the EU, not merely Iberjindal-finished product.
- **Indonesia's CBAM default is worse than India's:** base 8.69, 2026 mark-up 9.56 t/t.
- **Slab (CN 7218) carries no anti-dumping/CVD**, while Indian cold-rolled flat product carries approximately 56.7% combined AD+CVD, a live arbitrage favouring EU-side value addition.
- **India's CCTS draft (26 Jun 2026) covers 255 steel units**, intensity cuts 2.1–9.3% from FY26–27, a domestic-cost ledger separate from the EU border cost.
- **The first CBAM declaration for 2026 imports is due 30 Sep 2027**; 2026 tonnes can still switch default to verified if JSL is verified by then.

9. Competitor tools: Primetals, SSAB, Aperam, Outokumpu

File 07. Product documentation and UX walkthrough; Aperam and Outokumpu public carbon tools

	Primetals CO2 Calculator	SSAB EcoUpgraded	StainlessCarbon
Focus	carbon-steel route engineering	vehicle use-phase benefit (a different purpose)	stainless EAF-AOD production
Alloy chain (FeCr/NPI)	not modelled	not modelled	modelled per grade, with own-furnace vs merchant FeCr sourcing
Grid context	European defaults	n/a	India grid + costed captive coal load factor
Feasibility limits	none	n/a	grade-specific scrap caps; three optimiser tiers, each with disclosed costs
CBAM €/t	partial (ETS/CBAM cost indicators)	none	per installation, per t coil, 2026–30 (Jajpur 1.171 / Hisar 0.887)
Access	enterprise-gated; 30+ inputs	free app; 5–7 inputs	free web; 5 levers + grade incl. J4, 200-series

Stainless producers already publish carbon tools: Aperam's calculator compares footprints by origin country including India; Outokumpu issues certified per-batch footprints. Both are producer-side statements for their own steel. This tool's differentiators for an Indian mill: the alloy chain by grade and sourcing; India-specific factors (CEA grid, a costed captive-coal load factor); a per-installation CBAM figure on CBAM's actual boundary; an Indonesia-route comparison that flags misrouting; and grade J4, covering 200-series output no competitor tool models.

10. The scrap ceiling: tramp elements and sensor sorting

File 11 + file 01 §E

- In carbon steel, copper and tin cannot be oxidised out in the AOD and accumulate with scrap ratio, causing hot-shortness; this is the classic basis for carbon-steel scrap limits, not the basis for the caps below.
- Stainless caps come from scrap availability and chemistry match, grade by grade: nickel is a tramp element in Ni-free 430 (cap 60%), Mo-bearing scrap must be segregated for 316 (cap 75%), and in 200-series J4 (cap 70%) copper and manganese are intentional additions, not contaminants. India's scrap share is only 23–25% of steel input, and the global market is projected to swing to a deficit by 2030.
- The escape path is at the scrap yard: AI-driven sensor sorting upgrades tramp-laden scrap at approximately \$30–43/tCO₂, well below the 2030 CBAM price of €109.

11. Advanced levers (roadmap depth)

Files 09, 10, 13, 14, 15

- **Biochar in ferrochrome SAFs:** bamboo biochar is a near drop-in for met coke; 30% substitution abates 0.45–0.66 tCO₂/t FeCr at \$29 to \$45/tCO₂, aligned with the Ministry of Steel's 2024 roadmap.
- **Slag valorisation:** AOD slag substitutes cement clinker: Scope 1 mineralisation plus avoided Scope 3.
- **Logistics:** shifting inbound bulk to the Dedicated Freight Corridor saves approximately 40–67 kg CO₂e/tcs at near-zero cost.
- **Captive power value stacking:** outside the EU CBAM charge, so replacing it earns no direct CBAM saving; its value is CCTS compliance and the Scope 2 improvement, priced in section 6's load-factor step.
- **Three tiers, priced separately:** today (bio-LDO, RE 70%, biochar 20%, LF 0.55) averages \$37.7/tCO₂, reaches 2.30 t/t; stretch 2030 reaches 1.38; the theoretical floor reaches 0.73, a bound, not a plan.

12. Assumptions, limitations and source register

Sections 1, 2 and 6 carry the model's own honesty list in full; this register adds the items not already stated there.

- Intensities per tonne of crude steel unless stated per t coil; Scope 2 market-based.

- **FY26 NPI share (58%) is a plug, not a measurement:** it absorbs the unknown real grade mix, the FY26 Scope 3 method change, and the consumables estimate. Real NPI tonnage would expose the product-mix gap, the flagged next step.
- **Own-furnace ferrochrome's site attribution is a choice, not a measurement;** an alternative marginal-supply attribution gives a different, equally in-range FY24 result, depending on how JSL dispatches the captive plant when demand falls, a data ask.
- **CBAM precursor factors**, verified and default, are team assumptions or third-party compilations and need checking against the regulation's own Annex before external use.
- **Calibration:** seven constants fit seven FY26 targets exactly, identities not validation; real evidence is FY24 out-of-sample (within 0.65 to 0.8%), plausible implied fuel factors, and the worldstainless curve points.
- **Grade ranking differs from ISSF:** $430/304 = 0.62$ here vs approximately 0.95 in ISSF, because JSL's nickel is largely NPI, and own-furnace ferrochrome sits inside Scope 1+2.
- **Location-based Scope 2 does not reconcile:** disclosed figures imply a grid factor of 0.517, not 0.710, or a renewable share of approximately 27%, not 46.8%. Kept at 46.8% and flagged.
- **Net-to-JSL money** depends on Iberjindal's confirmed importer-of-record status (assumed), a 40% pass-through assumption, and an order-of-magnitude verification-cost estimate, all data asks.
- The FY26 Scope 3 drop is partly a methodology change (categories 10 and 12). Scrap caps are a supply-and-chemistry judgment; sensor sorting can raise them over time.

Primary sources

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Full research files with inline citations: the /research directory (research/22-canonical-model-v1.md, now v4, governs all figures; executable form website/model.js and website/business.js).